

ELEC4631 Lecture 9: Introduction to Linear Quadratic Regulator (LQR) Theory

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Control effort and cost

- We have now looked at pole placement as method for designing linear state-feedback control laws $u(t) = -Kx(t)$.
- However, this method on its own does not take into account the effort and cost involved in implementing $u(t)$.

- What cost could there be?
- In controlling physical systems the physical form of the signal $u(t)$ could be, for instance, a voltage or current. Large values of $u(t)$ mean large voltages or currents that could potentially damage the controlled system.
- Or, the signal may not be perfectly implementable due to actuator saturation, e.g., the current source cannot produce currents with magnitude larger than 1 mA.

- Some control designs may give reasonable control signals by chance rather than by design.
- However, there are design methods that explicitly take into account the cost involved with the control signal. They place a penalty on large control signals.
- We will now look at one of these methods for LTI state-space systems.

Notation

- We write $Q = Q^T$ to indicate that a (square) matrix Q is symmetric.
- We write $Q = Q^T > 0$ or simply $Q > 0$ (resp., $Q = Q^T \geq 0$ or simply $Q \geq 0$) to indicate that the symmetric matrix Q is positive definite (resp., positive semidefinite).
- For two symmetric matrices P and Q , $P > Q$ (resp., $P \geq Q$) means that $P - Q > 0$ (resp., $P - Q \geq 0$).

Linear quadratic regulator (LQR)

Consider an n -th order SISO LTI state-space system

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0,$$

where $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times 1}$, and $u(t) \in \mathbb{R}$.

We introduce the so-called linear quadratic (LQ) performance index or cost function:

$$J_{t_f}(x_0, u) = \int_0^{t_f} (x(t)^T Q x(t) + R u(t)^2) dt + x(t_f)^T \Gamma x(t_f).$$

where $0 < t_f < \infty$ is called the control horizon (later we will also allow $t_f = \infty$), $Q = Q^T \geq 0$ and $\Gamma = \Gamma^T \geq 0$, and R is a positive real scalar, $R > 0$.

- $J_{t_f}(x_0, u)$ is the sum of three terms.
- The first term $\int_0^{t_f} x(t)^T Q x(t) dt \geq 0$ places a penalty on the state $x(t)$.
- The second term $R \int_0^{t_f} u(t)^2 dt \geq 0$ places a penalty on the control signal $u(t)$.
- The term $x(t_f)^T \Gamma x(t_f)$ is a penalty on the final state $x(t_f)$ at time $t_f > 0$.
- Basically, Q , R , and Γ allows the assignment of different weights to $x(t)$, $u(t)$, and $x(t_f)$. They can be chosen to try to meet design objectives and to assign the relative significance of the three different terms in the sum.

- For instance, consider a system with two state variables, $Q = \text{diag}(q_1, q_2)$, with $q_1, q_2 > 0$ and $\Gamma = 0$ then

$$\begin{aligned} J_{t_f}(x_0, u) &= \int_0^{t_f} (x(t)^T Q x(t) + R u(t)^2) dt, \\ &= q_1 \int_0^{t_f} x_1(t)^2 dt + q_2 \int_0^{t_f} x_2(t)^2 dt \\ &\quad + R \int_0^{t_f} u(t)^2 dt. \end{aligned}$$

Setting $\Gamma = 0$ means that we do not care where the state will be at $t = t_f$. By varying q_1 , q_2 and R , different weights can be given to each integral in the sum. These weights are relative quantities, assigning weights cq_1 , cq_2 and cR , for a constant $c > 0$, is the same as assigning q_1 , q_2 and R . So in this case one can simply set $q_1 = 1$ and vary only q_2 and R (relative to q_1).

- Taking $q_1 = 1$,

$$J_{t_f}(x_0, u) = \int_0^{t_f} x_1(t)^2 dt + q_2 \int_0^{t_f} x_2(t)^2 dt + R \int_0^{t_f} u(t)^2 dt.$$

- Roughly speaking, setting $q_2 = 1$ and $R < 1$ means placing equal significance on the magnitude of the integral of x_1^2, x_2^2 and more significance (weight) on them over the magnitude of the integral of $u(t)^2$, setting $R > 1$ means giving more significance to the integral of $u(t)^2$ over that of x_1^2, x_2^2 , while setting $R = 1$ means putting equal significance on all terms.
- The designer can tweak the choice of Q, R, Γ until a satisfactory design is obtained.

- Example. If the system is second order, $n = 2$, and we want $x_1(t)$ to stay close to $x_2(t)$ on the interval $[0, t_f)$, and $x(t_f)$ to be close to 0, we can try setting

$$Q = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix}, \text{ and } \Gamma = \alpha I.$$

with $\alpha > 0$. Then $x(t)^T Q x(t) = (x_1(t) - x_2(t))^2$ and $x(t_f)^T \Gamma x(t_f) = \alpha \|x(t_f)\|^2$, so that

$$\begin{aligned} J_{t_f}(x_0, u) = & \int_0^{t_f} (x_1(t) - x_2(t))^2 dt \\ & + R \int_0^{t_f} u(t)^2 dt + \alpha \|x(t_f)\|^2. \end{aligned}$$

Choosing $\alpha > R > 1$ places more emphasis on the last term. Choosing $\alpha < R$ and $\alpha < 1$ places more emphasis on the first two terms.

- Example. If for the same second order system we want $x_1(t)$ to stay close to $x_2(t)$, and also for $x_2(t)$ to be close to 0, on $[0, t_f)$, but with more emphasis on the former, we can try setting

$$\begin{aligned} Q &= \begin{bmatrix} \kappa & -\kappa \\ 0 & \gamma \end{bmatrix}^T \begin{bmatrix} \kappa & -\kappa \\ 0 & \gamma \end{bmatrix}, \\ &= \begin{bmatrix} \kappa^2 & -\kappa^2 \\ -\kappa^2 & \kappa^2 + \gamma^2 \end{bmatrix}, \end{aligned}$$

with a choice $\kappa > \gamma > 0$. Then $x(t)^T Q x(t) = \kappa(x_1(t) - x_2(t))^2 + \gamma \|x_2(t)\|^2$.

- The linear quadratic regulator (LQR) with horizon t_f is a piecewise-continuous control law u that minimises the performance index $J_{t_f}(x_0, u)$. That is, an LQR control signal u_{opt} satisfies

$$J_{t_f}(x_0, u_{\text{opt}}) \leq J_{t_f}(x_0, u),$$

over all possible piecewise-continuous control signals u .

- Finding u_{opt} that minimises the performance index $J_{t_f}(x_0, u)$ is an optimisation problem. LQR design is part of optimal control theory – the theory of finding optimal controllers that optimise a desired performance index.

- It is not an ordinary optimisation because the optimisation is over a function space, i.e., the space of all piecewise continuous functions on $[0, t_f]$, not over a Euclidean space. That is, we need to optimise over functions rather than over vectors in some Euclidean space $\mathbb{R}^n$.
- When $u = u_{\text{opt}}$ then the performance index is minimised and we write $J_{t_f}(x_0, u_{\text{opt}}) = J_{t_f, \text{opt}}(x_0)$, where the optimal performance index $J_{t_f, \text{opt}}(x_0)$ depends on x_0 and t_f (as does u).

- Aside: The most general form of the finite horizon LQR problem for SISO LTI state-space systems, which we do not consider here, is:

$$\begin{aligned} J_{t_f}(x_0, u) \\ = \int_0^{t_f} (x^T Q x + x^T N u(t) + R u(t)^2) dt \\ + x(t_f)^T \Gamma x(t_f), \end{aligned}$$

where $N \in \mathbb{R}^{n \times 1}$ satisfies $Q - R^{-1} N N^T \geq 0$ to ensure that $J_{t_f}(x_0, u) \geq 0$.

- Optimisation over function spaces is the subject of calculus of variations and dynamic programming.
- Dynamic programming is a versatile method developed by Richard Bellman, a mathematician, in the early 50's. The method has natural applications in control theory and various other fields.
- Dynamic programming can solve a large class of optimisation problems called Markov Decision Problems (MDPs) – the LQR problem belongs in this class. It also includes some problems in machine learning, such as finding algorithms for playing games, e.g., TD-Gammon, a reinforcement learning algorithm for playing backgammon.

- Optimal control theory, that use methods from the calculus of variations and dynamic programming, is a deep subject that in itself can be the subject of an entire course. We do not cover it here. However, we shall contend ourselves with knowing the solution of the LQR problem given by these methods.
- For those interested in knowing more about optimal control theory and dynamic programming, a recommended classic undergraduate text is: W. R. Brogan, Modern Control Theory, 3rd edition, Prentice-Hall, 1991.

Solution of LQR with a finite horizon

- The results of LQR theory are very impressive, also due to Rudolf Kalman. It turns out, surprisingly, that the solution to LQR is a linear time-varying feedback control law

$$u_{\text{opt}}(t) = -K(t)x(t),$$

where $K(t) \in \mathbb{R}^{1 \times n}$ is a time-varying feedback gain given by $K(t) = R^{-1}B^T P(t)$, and $P(t)$ is a symmetric $n \times n$ matrix (i.e., $P(t) = P(t)^T$) for each t and solves the so-called Riccati differential equation (RDE)

$$\begin{aligned} -\dot{P}(t) &= A^T P(t) + P(t)A \\ &\quad - P(t)BR^{-1}B^T P(t) + Q, \end{aligned}$$

with terminal condition $P(t_f) = \Gamma$.

- The RDE

$$\begin{aligned} -\dot{P}(t) = & A^T P(t) + P(t)A \\ & -P(t)BR^{-1}B^T P(t) + Q, \end{aligned}$$

needs to be solved backwards in time starting from the final time $t = t_f$ using the terminal condition $P(t_f) = \Gamma$.

- Solving the RDE, the optimal performance index is:

$$J_{t_f, \text{opt}}(x_0) = J_{t_f}(x_0, u_{\text{opt}}) = x_0^T P(0) x_0.$$

- Some properties of the solution of the RDE:
 1. There is a unique solution $P(t)$ of the RDE on the interval $(-\infty, t_f]$.
 2. $P(t) = P(t)^T \geq 0$.
 3. If $\Gamma = 0$, $P(t)$ is non-increasing: for any $t_1, t_2 \in (-\infty, t_f]$, $P(t_1) \geq P(t_2)$ if $t_1 \leq t_2$. (Recall that $P(t_1) \geq P(t_2)$ means that $P(t_1) - P(t_2) \geq 0$).

- The most important feature of the solution is that the Riccati differential equation, and therefore $P(t)$, does not depend on $x(t)$ and $u(t)$.

- Hence, $P(t)$ can be pre-determined “off-line” by solving the RDE and then used to compute $u_{\text{opt}}(t) = -R^{-1}B^T P(t)x(t)$ in a real-time implementation.

LQR with an infinite horizon

- Often, it is desired to maintain a certain performance over all times t rather than just over a finite interval $[0, t_f]$. That is, it is meaningful to consider letting $t_f \rightarrow \infty$.
- It is therefore of practical relevance to consider a performance index over an infinite-horizon of the form:

$$J_\infty(x_0, u) = \int_0^\infty (x(t)^T Q x(t) + R u(t)^2) dt.$$

It may be possible that $J_\infty(x_0, u) = \infty$ for some choices of input functions u .

- Aside: The most general form of the infinite horizon LQR problem for SISO LTI state-space systems, which we do not consider here, is:

$$\begin{aligned} J_{\infty}(x_0, u) \\ = \int_0^{\infty} (x^T Q x + x^T N u(t) + R u(t)^2) dt \end{aligned}$$

where $N \in \mathbb{R}^{n \times 1}$ satisfies $Q - R^{-1} N N^T \geq 0$ to ensure that $J_{\infty}(x_0, u) \geq 0$.

- We will now need the following observation: If $Q = Q^T \in \mathbb{R}^{n \times n}$ and $Q \geq 0$ then there is a unique matrix $A = A^T \in \mathbb{R}^{n \times n}$ such that $A \geq 0$ and $Q = A^2$. We call A the square root of Q and write $A = Q^{1/2}$.
- The proof is fairly straightforward:

$$Q = V \operatorname{diag}(d_1, d_2, \dots, d_n) V^T,$$

with $d_j \geq 0$ for $j = 1, 2, \dots, n$ for some real orthogonal matrix V . Define

$A = V \operatorname{diag}(\sqrt{d_1}, \sqrt{d_2}, \dots, \sqrt{d_n}) V^T$. Then clearly $A = A^T$, $A \geq 0$, and $A^2 = Q$. (Continued to next page).

- A is unique and is of the form above. To see this, first note that since any square root A is symmetric and nonnegative definite it has n real nonnegative eigenvalues and n mutually orthonormal eigenvectors, say, $v_1, v_2, \dots, v_n$. For any eigenvector v_k corresponding to an eigenvalue λ_k of A we have that $Qv_k = A^2v_k = \lambda_k^2v_k$. Therefore, λ_k^2 and v_k for $k = 1, 2, \dots, n$ are the eigenvalues and eigenvectors of Q . That is, Q and A share an identical complete set of n eigenvectors. Thus A is always of the form

$$A = V \text{diag}(\sqrt{d_1}, \sqrt{d_2}, \dots, \sqrt{d_n}) V^T$$

with $d_1, d_2, \dots, d_n$ the eigenvalues of Q and $V = [v_1 \ v_2 \ \dots \ v_n]$ with $v_1, v_2, \dots, v_n$ mutually orthonormal eigenvectors of Q . This defines A uniquely.

- We also need the notion of a controllable and observable pair of matrices.
- A pair of matrices (A, B) with $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$ is said to be controllable or is a controllable pair if the controllability matrix

$$\mathcal{C} = \begin{bmatrix} B & AB & \dots & A^{n-1}B \end{bmatrix}$$

has rank n .

- A pair of matrices (C, A) with $C \in \mathbb{R}^{m \times n}$ and $A \in \mathbb{R}^{n \times n}$ is said to be observable or is an observable pair if the observability matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix},$$

has rank n .

- Useful note: (A, B) is a controllable pair if and only if (B^T, A^T) is an observable pair.

- If (A, B) is a controllable pair and $(Q^{1/2}, A)$ is an observable pair then $J_\infty(x_0, u)$ can be thought of as the limit of a finite horizon performance index $J_{t_f}(x_0, u)$

$$J_{t_f}(x_0, u) = \int_0^{t_f} (x(t)^T Q x(t) + R u(t)^2) dt,$$

as $t_f \rightarrow \infty$.

- Here the controllability of (A, B) guarantees that there is always a choice of u which makes

$$\lim_{t_f \rightarrow \infty} J_{t_f}(x_0, u) < \infty,$$

for any x_0 , implying that $J_\infty(x_0, u) < \infty$.

- Observability of $(Q^{1/2}, A)$ ensures that

$$\lim_{t_f \rightarrow \infty} J_{t_f}(x_0, u) = \infty$$

whenever $\lim_{t_f \rightarrow \infty} \|x(t_f)\| = \infty$.

- Example 1. Consider a second order system with

$$\dot{x}(t) = x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t),$$

and suppose that $Q = I_{2 \times 2}$. (A, B) is not controllable and there is no u such that

$$\lim_{t_f \rightarrow \infty} J_{t_f}(x_0, u) < \infty$$

for all $x_0 = (x_{01}, x_{02})^T$ such that $x_{01} \neq 0$.

- Example 2. Consider the same second order system as above with $Q = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. $(Q^{1/2}, A)$ is not observable and we can have a u such that

$$\lim_{t_f \rightarrow \infty} J_{t_f}(x_0, u) < \infty$$

although $\lim_{t_f \rightarrow \infty} |x_1(t_f)| = \infty$ (thus also $\lim_{t_f \rightarrow \infty} \|x(t_f)\| = \infty$).

- Example 3. Consider a second order system with

$$\dot{x}(t) = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} x(t) + \begin{bmatrix} 2 \\ 1 \end{bmatrix} u(t),$$

and take the infinite horizon cost to be

$$J_{\infty}(x_0, u) = \int_0^{\infty} ((x_1(t) - x_2(t))^2 + ru(t)^2) dt$$

for some $r > 0$.

- In this case $Q = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ and $(Q^{1/2}, A)$ is not observable/an observable pair.
- It can be shown that under an optimal control law that minimises J_{∞} , $\|x(t)\| \rightarrow \infty$ while $x_1(t) - x_2(t) \rightarrow 0$ as $t \rightarrow \infty$ from some initial states x_0 . Thus the closed-loop system under the optimal control law is unstable.

Under the assumptions that (A, B) is controllable and $(Q^{1/2}, A)$ is observable we have

1. Let $P_{t_f}(t)$ be the solution of the RDE

$$\begin{aligned} -\dot{P}_{t_f}(t) &= A^T P_{t_f}(t) + P_{t_f}(t) A \\ &\quad - P_{t_f}(t) B R^{-1} B^T P_{t_f}(t) + Q, \end{aligned}$$

on $(-\infty, t_f]$ with $P_{t_f}(t_f) = 0$. Then

$$\lim_{t_f \rightarrow \infty} P_{t_f}(0) = P,$$

where $P = P^T > 0$ satisfies the continuous-time algebraic Riccati equation (CARE):

$$A^T P + P A - P B R^{-1} B^T P + Q = 0.$$

There exists only one solution P satisfying $P = P^T > 0$.

2. Optimal LQR is the linear feedback law $u_{\text{opt}}(t) = -Kx(t)$ with $K = R^{-1}B^T P$.

3. The optimal performance index is

$$J_{\infty, \text{opt}}(x_0) = J_{\infty}(x_0, u_{\text{opt}}) = x_0^T P x_0.$$

4. The controller is stabilising: $A_{\text{cl}} = A - BK = A - BR^{-1}B^T P$ has all its eigenvalues in the LHP. Thus the closed-loop system with the state feedback $u_{\text{opt}}(t)$ is asymptotically stable.
- Thus, LQR theory essentially provides a method for designing optimal stabilising controllers that force $\|x(t)\| \rightarrow 0$ as $t \rightarrow \infty$ but balanced with a penalty on the control effort needed to achieve the stabilisation.

Example 1: 1D acceleration control

- (From R. Kwong, Lecture notes ECE410, University of Toronto, 2008). A vehicle is moving along a line with acceleration control. The state is $\xi(t) = (x, \dot{x})^T$, where x is the position. The state-space equation is:

$$\dot{\xi}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \xi(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t).$$

We want to control the vehicle so that $x(t) \rightarrow 0$ as $t \rightarrow \infty$.

- We consider the performance index

$$J_{\infty}(\xi_0, u) = \int_0^{\infty} (\xi(t)^T Q \xi(t) + r u(t)^2) dt,$$

with $r > 0$ and $Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

- Here $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and $Q^{1/2} = Q$.

It is easy to check that (A, B) and $(Q^{1/2}, A)$ are controllable and observable pairs, respectively. Thus the optimal LQ solution exists and will be given by $u_{\text{opt}}(t) = -R^{-1}B^T P x(t)$. We need to find the solution $P = P^T \in \mathbb{R}^{2 \times 2}$ to the CARE

$$A^T P + P A - P B R^{-1} B^T P + Q = 0.$$

- Let $P = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix}$ then the CARE results in the set of equations: (i) $-p_2^2/r + 1 = 0$, (ii) $p_1 - p_2 p_3/r = 0$, and (iii) $2p_2 - p_3^2/r = 0$.

- Eq. (i) gives $p_2 = \pm\sqrt{r}$, eq. (iii) gives $p_3 = \pm(2rp_2)^{1/2} \Rightarrow p_2 = \sqrt{r}$. Since $P > 0$ we must have $p_3 > 0$, thus $p_3 = \sqrt{2}r^{3/4}$. Finally, from (ii) we obtain $p_1 = p_2 p_3/r = \sqrt{2}r^{1/4}$. So, $P = \begin{bmatrix} \sqrt{2}r^{1/4} & \sqrt{r} \\ \sqrt{r} & \sqrt{2}r^{3/4} \end{bmatrix}$ and $P > 0$.

- The state equation for the closed-loop system is:

$$\begin{aligned}
\dot{\xi}(t) &= (A - BR^{-1}B^T P)\xi(t), \\
&= \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{r} \begin{bmatrix} \sqrt{r} & \sqrt{2}r^{3/4} \end{bmatrix} \right) \xi(t), \\
&= \begin{bmatrix} 0 & 1 \\ -r^{-1/2} & -\sqrt{2}r^{-1/4} \end{bmatrix} \xi(t).
\end{aligned}$$

- The characteristic polynomial of

$$A_{\text{cl}} = A - BR^{-1}B^T P$$

is $c_{A_{\text{cl}}}(s) = s^2 + \sqrt{2}r^{-1/4}s + r^{-1/2}$. This can be written as $c_{A_{\text{cl}}}(s) = s^2 + 2\zeta\omega_n s + \omega_n^2$, where $\omega_n = r^{-1/4}$ and $\zeta = 1/\sqrt{2}$ are the natural frequency and damping ratio of the second order system, respectively. The two roots of $c_{A_{\text{cl}}}(s)$ are in the LHP.

- For smaller r (less penalty on the control, larger control signals) ω_n is larger implying a faster response (closed-loop poles in the LHP further away from the imaginary axis), while for larger r (more penalty on the control, smaller control signals) ω_n is smaller implying a slower response (closed-loop poles in the LHP closer to the imaginary axis).
- The results of the calculations for this basic example agree with what is expected by intuition.

Example 2: Servomotor control

- (From R. Kwong, Lecture notes ECE410, University of Toronto, 2008). Let $\xi(t)$ be as in Example 1. The state-space equation for a servomotor system with velocity feedback is:

$$\dot{\xi}(t) = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t).$$

We want to control the servomotor so that $x(t) \rightarrow 0$ as $t \rightarrow \infty$.

- We consider the performance index

$$J_{\infty}(\xi_0, u) = \int_0^{\infty} (\xi(t)^T Q \xi(t) + r u(t)^2) dt,$$

with $r > 0$ and $Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, as in Example 1.

- Here $A = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and $Q^{1/2} = Q$.

Again, it is easy to check that (A, B) and $(Q^{1/2}, A)$ are controllable and observable pairs, respectively. Thus the optimal LQ solution exists and will be given by $u_{\text{opt}}(t) = -R^{-1}B^T P x(t)$. We again seek the solution $P = P^T \in \mathbb{R}^{2 \times 2}$ to the CARE

$$A^T P + P A - P B R^{-1} B^T P + Q = 0.$$

- As before, let $P = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix}$ then the CARE results in the set of equations: (i) $-p_2^2/r + 1 = 0$, (ii) $p_1 - p_2 - p_2 p_3/r = 0$, and (iii) $2(p_2 - p_3) - p_3^2/r = 0$.

- Solving equations (i)-(iii) in the same manner as before gives the solution $p_2 = \sqrt{r}$, $p_3 = r\sqrt{1 + 2r^{-1/2}} - r$, and $p_1 = \sqrt{r + 2r^{1/2}}$. So,

$$P = \begin{bmatrix} \sqrt{r + 2r^{1/2}} & \sqrt{r} \\ \sqrt{r} & r\sqrt{1 + 2r^{-1/2}} - r \end{bmatrix} > 0.$$

- The state equation for the closed-loop system is:

$$\begin{aligned}
\dot{\xi}(t) &= (A - BR^{-1}B^TP)\xi(t), \\
&= \left(\begin{bmatrix} 0 & 1 \\ p_2/r & -1 - p_3/r \end{bmatrix} \right) \xi(t). \\
&= \begin{bmatrix} 0 & 1 \\ -r^{-1/2} & -\sqrt{1 + 2r^{-1/2}} \end{bmatrix} \xi(t).
\end{aligned}$$

- The characteristic polynomial of

$$A_{cl} = A - BR^{-1}B^TP$$

is $c_{A_{cl}}(s) = s^2 + \sqrt{1 + 2r^{-1/2}}s + r^{-1/2}$. The poles are at $s = \frac{1}{2}(-\sqrt{1 + 2r^{-1/2}} \pm \sqrt{1 - 2r^{-1/2}})$. Both poles are in the LHP.

- Again, it can be seen that for smaller r (less penalty on the control, larger control signals) both closed-loop poles move away from the imaginary axis in the LHP implying a faster response, while for larger r (more penalty on the control, smaller control signals) both closed-loop poles move closer to the origin in the LHP implying a slower response.